

AI MENTAL HEALTH ANA – 1

Report on Dataset Profiling and Summary Stats based on the Dataset Adult Psychiatric Morbidity Survey 2014.

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1. Introduction

In today's time, mental health is becoming an important topic because it affects people's daily life and productivity. The Adult Psychiatric Morbidity Survey (APMS) 2014 provides useful data about mental health conditions among adults. The main aim of this analysis is to understand the dataset and perform basic statistical analysis on it. In this report, I have analyzed how mental health severity is distributed across different categories using simple statistical methods and graphs.

2. Dataset Description

The dataset used in this analysis is taken from the Adult Psychiatric Morbidity Survey (APMS) 2014. It contains percentage-based values that represent the severity of common mental disorders among adults.

To keep the analysis simple and easy to understand, I have used only the “**All Adults**” section of the dataset. This section gives an overall picture without separating data into men and women.

The dataset is divided into different categories based on severity levels:

- **0–5** → No or very low symptoms
- **6–11** → Mild symptoms
- **Under 12** → Below clinical threshold
- **12–17** → Moderate symptoms
- **18 or more** → Severe symptoms
- **12 or more** → Clinically significant cases

Each category contains percentage values across different age groups.

	Category	76.5	72.3	69.7	72.1	69.9	81.3	81.9	74
13	0-5	67.1	66.3	66.4	65.3	66.3	76.6	77.6	68.6
14	6-11	15.5	16.5	15.7	16.7	16.7	13.2	14.3	15.7
15	Under 12	82.7	82.8	82.1	82.0	82.9	89.8	91.9	84.3
16	12-17	7.8	9.0	8.1	8.2	7.8	5.9	4.7	7.6
17	18 or more	9.5	8.2	9.8	9.8	9.2	4.2	3.3	8.1
18	12 or more	17.3	17.2	17.9	18.0	17.1	10.2	8.1	15.7

3. Data Types Analysis

Before starting any analysis, it is important to understand the type of data present in the dataset.

In this dataset:

- The **Category column** is of object type (text), which represents different mental health severity levels.
- All other columns are of float type (numeric), which represent percentage values.

This means that the dataset contains both categorical and numerical data. The numerical columns can be used for statistical calculations such as mean, minimum, and maximum values. Understanding data types helps in choosing the correct analysis methods and avoiding errors during computation.

`df_final.dtypes`

	0
Category	object
76.5	float64
72.3	float64
69.7	float64
72.1	float64
69.9	float64
81.3	float64
81.9	float64
74	float64

dtype: object

4. Missing Values Analysis

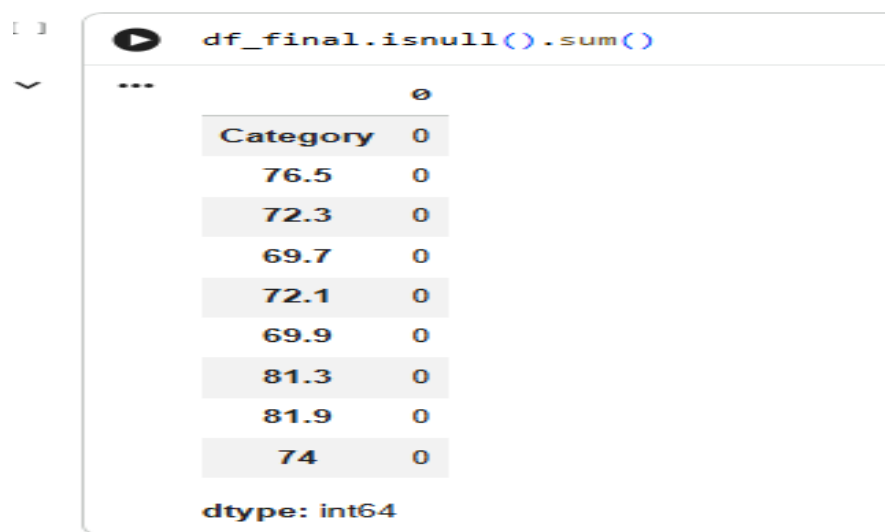
Missing values can affect the accuracy of the results, so it is important to check whether the dataset contains any missing data.

After cleaning the dataset, I checked for missing values using a simple function.

The result shows that:

- There are no missing values in the dataset
- All columns contain complete data

This indicates that the dataset is clean and reliable for further analysis.



```
[ ] df_final.isnull().sum()
```

	0
Category	0
76.5	0
72.3	0
69.7	0
72.1	0
69.9	0
81.3	0
81.9	0
74	0

dtype: int64

5. Descriptive Statistics

Descriptive statistics help in understanding the overall distribution of the dataset.

Using statistical functions, I calculated:

- Mean (average value)
- Minimum value
- Maximum value
- Standard deviation
- Quartiles

From the analysis, the following observations can be made:

- The **“Under 12” category** has the highest values, which shows that most individuals fall below the clinical threshold.
- The **“0–5 category”** has moderate values, indicating people with minimal symptoms.
- The **“12–17” and “18 or more” categories** have lower values, which means fewer people fall into moderate or severe conditions.

- The values gradually decrease as severity increases.

These statistics give a clear idea about how mental health conditions are distributed among adults.

```
df_final.describe()
```

	76.5	72.3	69.7	72.1	69.9	81.3	81.9	74
count	6.000000	6.000000	6.000000	6.000000	6.000000	6.000000	6.000000	6.000000
mean	33.316667	33.333333	33.333333	33.333333	33.333333	33.316667	33.316667	33.333333
std	32.779043	32.561306	32.283907	31.898568	32.613535	38.993149	40.275174	33.947410
min	7.800000	8.200000	8.100000	8.200000	7.800000	4.200000	3.300000	7.600000
25%	11.000000	10.875000	11.275000	11.525000	11.075000	6.975000	5.550000	10.000000
50%	16.400000	16.850000	16.800000	17.350000	16.900000	11.700000	11.200000	15.700000
75%	54.650000	54.025000	54.275000	53.475000	54.000000	60.750000	61.775000	55.375000
max	82.700000	82.800000	82.100000	82.000000	82.900000	89.800000	91.900000	84.300000

7. Outlier Detection

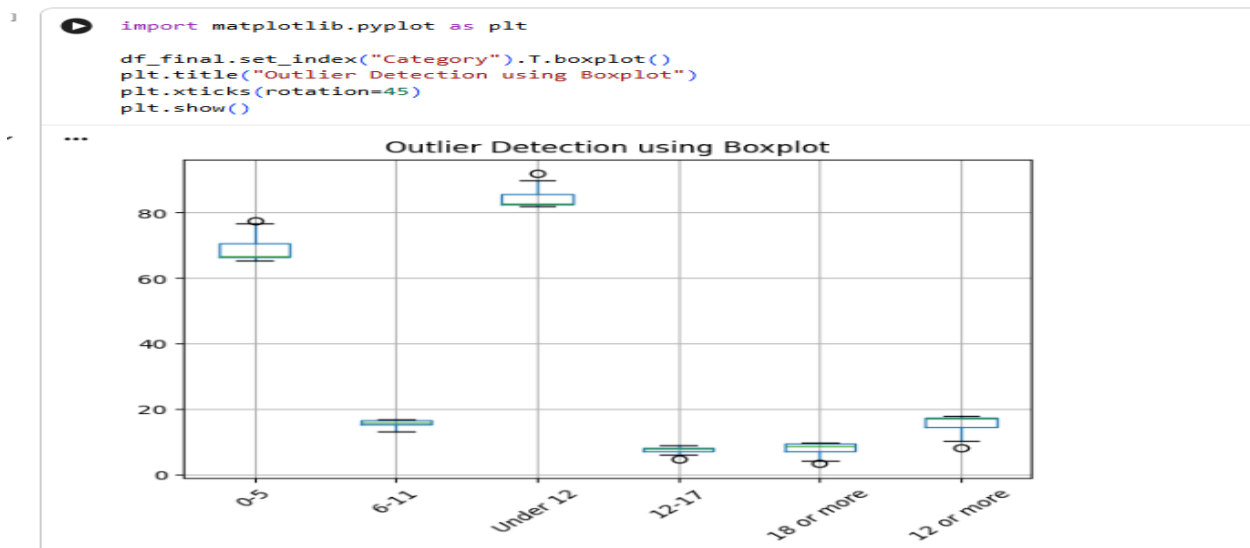
Outliers are values that are very different from the rest of the data. These values can sometimes affect the results of analysis.

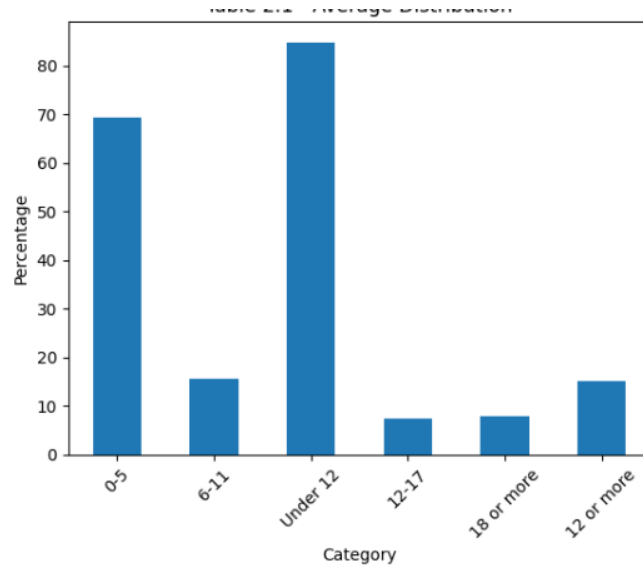
To detect outliers, I used a boxplot.

The results show that:

- There are no major outliers in the dataset
- Some small variations are present, but they are within a normal range

This means that the dataset is stable and does not contain abnormal values.





8. Conclusion

From this analysis, it can be concluded that:

- Most individuals fall under lower mental health severity categories
- Severe mental health conditions are less common
- The dataset is clean and does not contain missing values or major outliers

The use of descriptive statistics and visualization helped in understanding the dataset clearly.

Overall, this analysis provides a simple and clear view of mental health distribution among adults based on APMS 2014 data.